

SCSS NOTES

What is SCSS

SCSS (Sassy CSS) is a **more powerful version of CSS** that makes writing styles **easier and more organized**.

Browsers **cannot read SCSS directly**, so we **compile it into CSS** before using it.

SCSS adds:

1. **Variables** – Store values (like colors or spacing) once and reuse everywhere.

```
2. $primary: #2563eb;
```

```
3. button { background: $primary; }
```

4. **Nesting** – Write child styles inside parent blocks for better readability.

```
5. nav {
```

```
6.   background: #fff;
```

```
7.   a {
```

```
8.     color: blue;
```

```
9.     &:hover { color: red; }
```

```
10.  }
```

```
11. }
```

12. **Mixins** – Create reusable groups of styles.

```
13. @mixin card {
```

```
14.   border-radius: 10px;
```

```
15.   box-shadow: 0 5px 10px rgba(0,0,0,0.1);
```

```
16. }
```

```
17.
```

```
18. .card { @include card; }
```

19. **Import & Partial Files** – Organize large CSS into smaller files.

```
20. @import "buttons";
```

```
21. @import "header";
```

22. **Math & Logic** – Perform calculations in your styles.

```
23. $base-size: 16px;
```

```
24. h1 { font-size: $base-size * 2; }
```

Feature	CSS	SCSS
Variables	--color: red; (runtime)	\$color: red; (compile-time)
Nesting	❌ Not supported	✅ Fully supported
Mixins	❌ Not available	✅ Reusable with @mixin & @include
Math / Logic	❌ Limited	✅ Supports calculations & conditions
Readability	Flat, repetitive	Organized, readable
Compilation	Directly in browser	Needs to be compiled into CSS
Best For	Small projects	Large, professional projects

As per the example:

1. SCSS Variables

- `$radius`, `$shadow` are pre-processed — replaced before compilation.
- You can change one variable, and all dependent styles update automatically.

2. Mixins

- `@mixin card-style` groups reusable properties.
- Use with `@include card-style;` — saves repetition.

3. Nesting

- Inside `.header { button { ... } }` means “button inside header”.
- Cleaner than writing `.header button { ... }` every time.

4. CSS Variables for Themes

- `--bg`, `--text`, etc. are runtime variables — they change when theme switches.
- `[data-theme="dark"]` changes their values live.

5. Compilation

- Browser reads **only CSS**.
- SCSS must be compiled using the `sass` command.